

Experiment 7

EDA – Quantitative and Qualitative Analysis.

Program:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
tips = sns.load_dataset('tips')
print(tips.head())
sns.displot(tips.total_bill, kde=True)
sns.displot(tips.total_bill, kde=False)
sns.jointplot(x=tips.tip, y=tips.total_bill)
sns.jointplot(x=tips.tip, y=tips.total_bill, kind="reg")
sns.jointplot(x=tips.tip, y=tips.total_bill, kind="hex")
sns.pairplot(tips)
print(tips.time.value_counts())
sns.pairplot(tips, hue='time')
sns.pairplot(tips, hue='day')
sns.heatmap(tips.corr(numeric_only=True), annot=True)
sns.boxplot(x=tips.total_bill)
sns.boxplot(x=tips.tip)
sns.countplot(x=tips.day)
sns.countplot(x=tips.sex)
tips.sex.value_counts().plot(kind='pie')
tips.sex.value_counts().plot(kind='bar')
sns.countplot(x=tips[tips.time=='Dinner']['day'])
print("All plots generated successfully.")
```

Output

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

time

time

Dinner 176

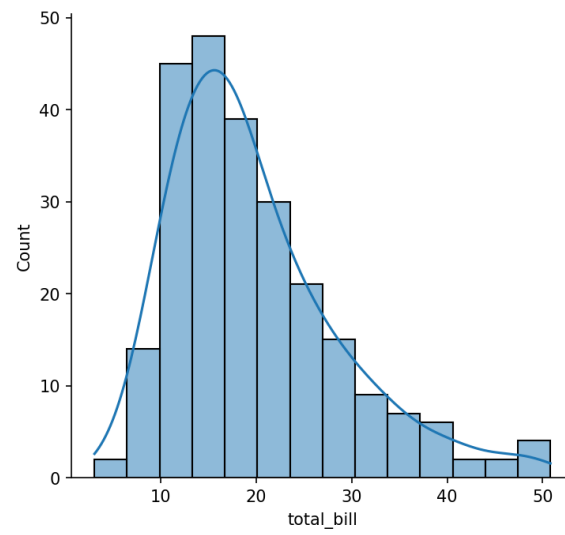
Lunch 68

Name: count, dtype: int64

All plots generated successfully.

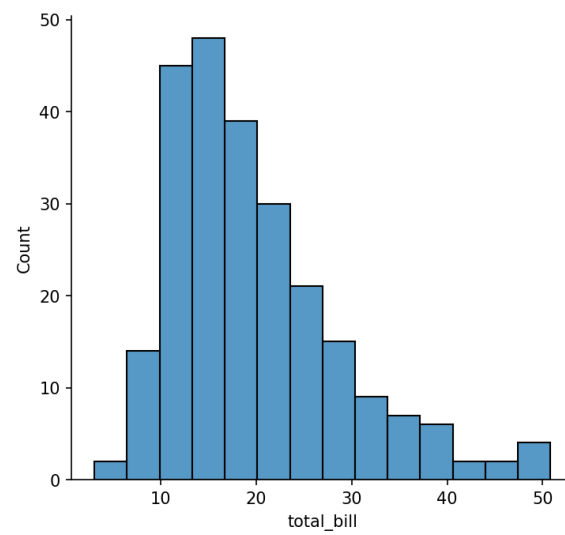
Output

Distribution of Total Bill (KDE)

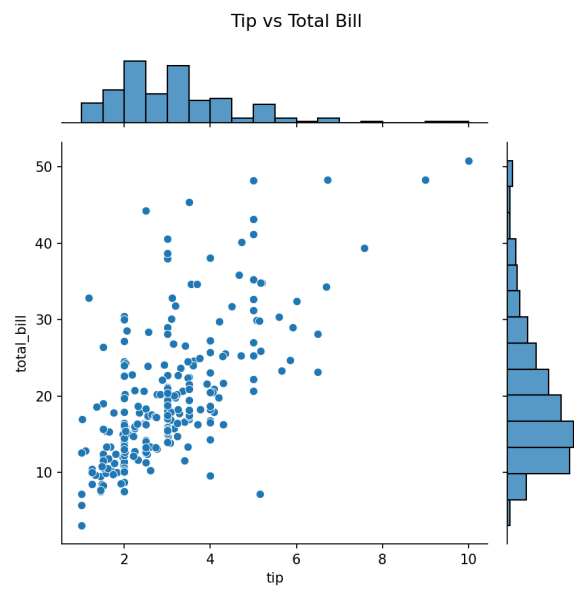


Output

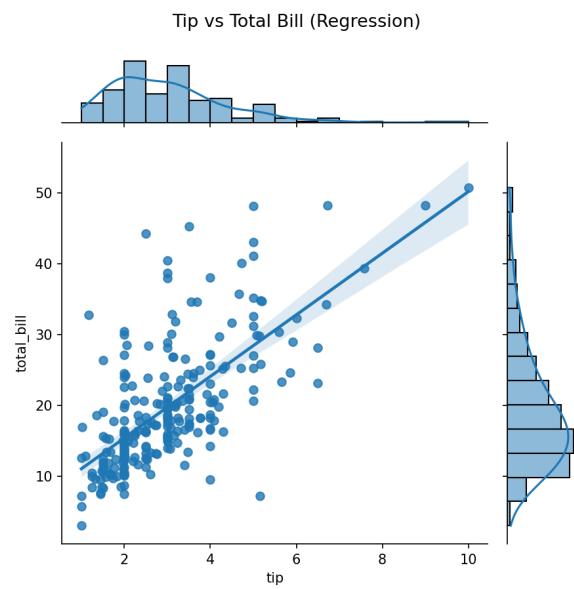
Distribution of Total Bill



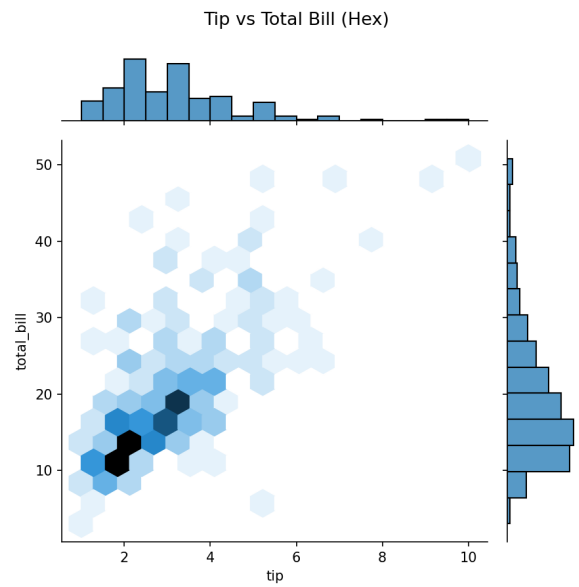
Output



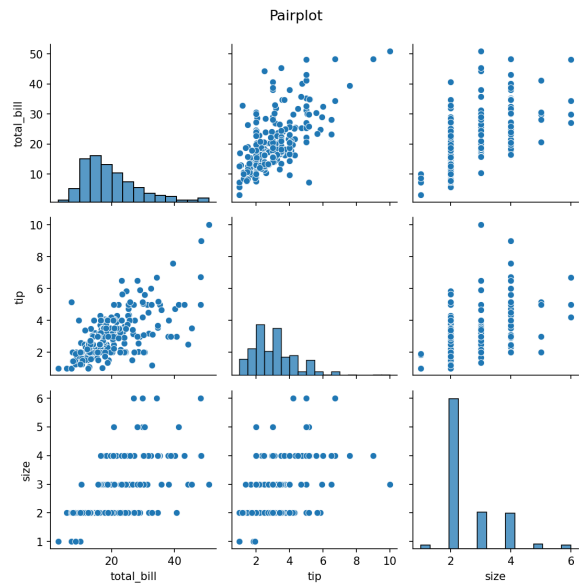
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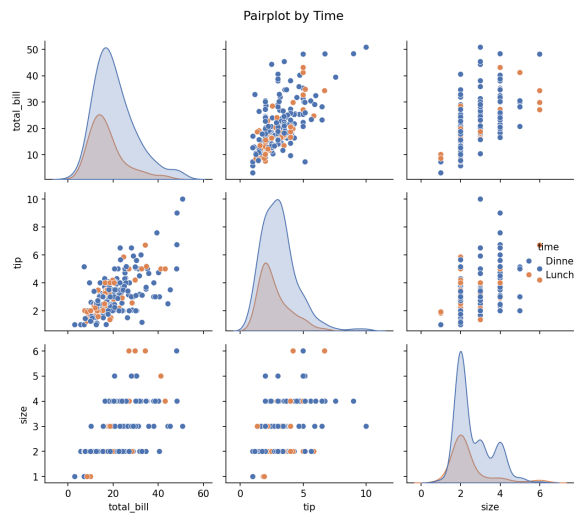
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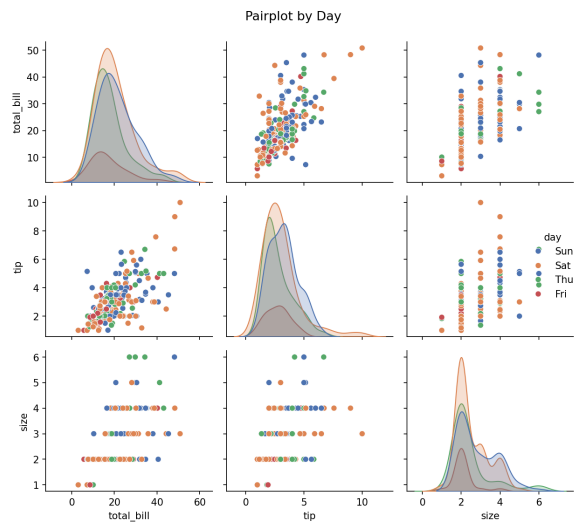
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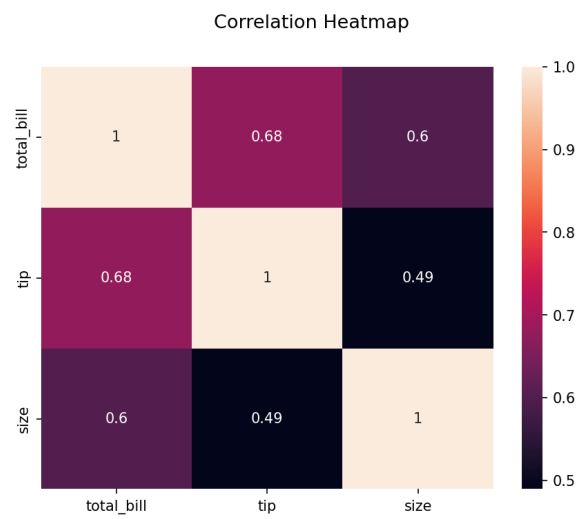
Output



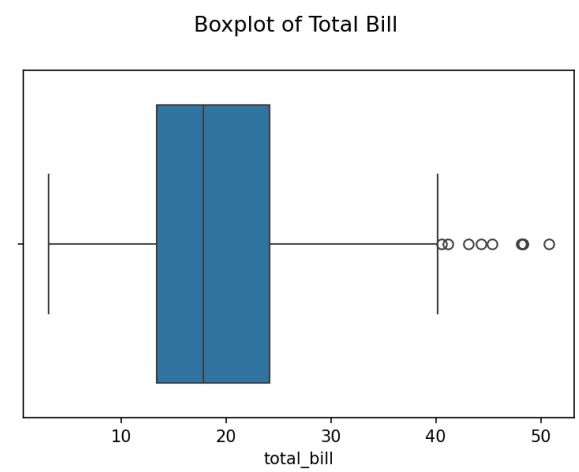
Output



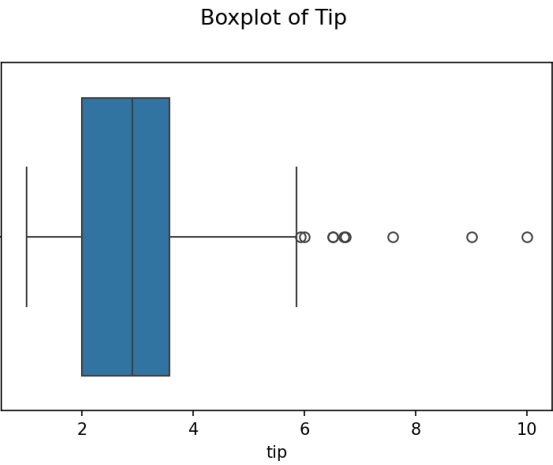
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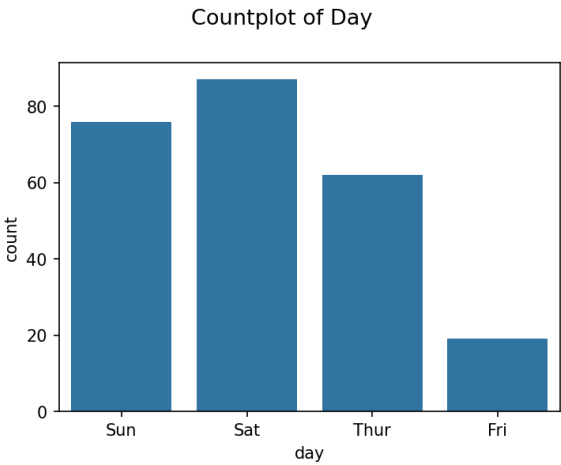
Output



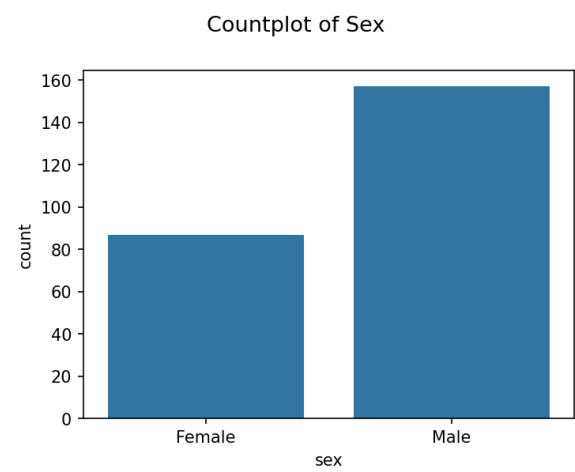
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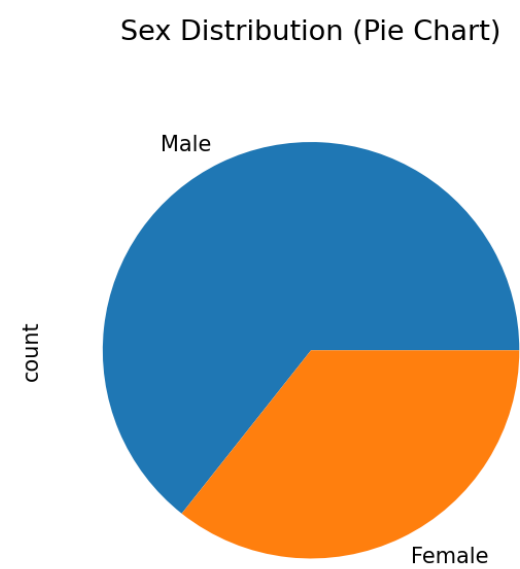
Output



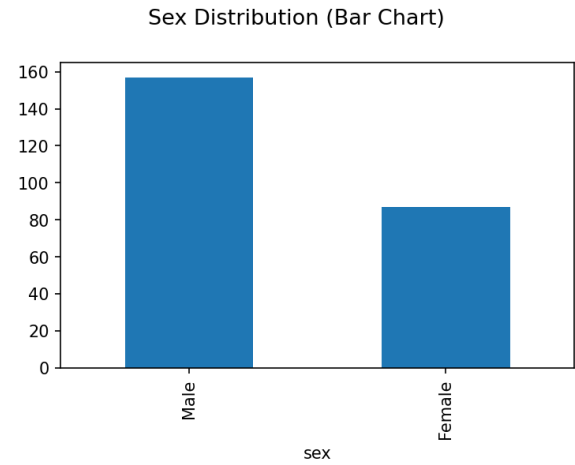
Output



Output



Output



Output

